

HIMEJI, Japan

WMO: 477690

Lat: 34.8394N

Lon: 134.6708E

Elev: 40

StdP: 100.85

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-1.2	-8.2	1.9	2.6	-6.8	2.1	3.1	9.5	5.4	8.5	5.3	2.6	330	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.6	33.6	25.7	32.6	25.4	31.5	25.0	26.6	31.6	26.1	30.7	25.6	29.9	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.8	24.8	19.9	28.5	24.4	19.4	28.2	83.3	31.6	81.1	30.9	79.2	30.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.0	6.0	-4.1	35.1	1.2	0.8	-4.9	35.7	-5.6	36.2	-6.2	36.6	-7.1	37.2		
			DB													
			WB	-5.1	27.4	1.3	0.5	-6.0	27.8	-6.7	28.1	-7.4	28.4	-8.4	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	5.2	8.3	13.5	18.5	22.5	26.7	28.2	24.1	18.2	12.2	6.9	(6)
		DBStd	8.52	2.12	2.59	2.99	3.23	2.29	2.17	2.14	1.76	2.84	2.88	3.11	2.67	
		HDD10.0	499	168	137	71	8	0	0	0	0	0	13	102		
		HDD18.3	1854	426	368	310	146	26	1	0	1	37	186	354		
		CDD10.0	2618	1	3	19	113	264	375	517	563	424	256	78	6	
		CDD18.3	930	0	0	0	2	32	126	258	304	174	34	0	0	
		CDH23.3	8292	0	0	0	9	147	691	2486	3457	1364	138	1	0	
		CDH26.7	2837	0	0	0	0	9	112	872	1447	388	9	0	0	
(14)	Wind	WSAvg	2.7	2.8	2.7	2.9	2.8	2.7	2.5	2.7	2.9	2.8	2.5	2.5	2.7	(14)
(15)	Precipitation	PrecAvg	1259	50	60	97	98	144	150	178	102	152	106	72	50	(15)
(16)		PrecMax	1770	123	133	151	204	312	276	378	247	403	266	169	89	(16)
(17)		PrecMin	778	14	18	42	24	56	56	21	9	42	21	11	11	(17)
(18)		PrecStd	256	26	30	26	44	62	56	112	73	86	58	46	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	13.6	16.3	19.7	24.5	27.9	31.1	34.3	35.1	32.8	27.9	22.1	16.7		(19)
(20)		0.4% MCWB	9.1	12.6	13.2	16.2	18.8	23.9	26.2	25.9	25.2	21.8	16.3	12.5		(20)
(21)		2% DB	11.6	13.7	17.5	22.5	26.2	29.1	33.1	34.0	31.4	26.1	20.5	14.8		(21)
(22)		2% MCWB	7.6	9.8	12.2	15.2	18.1	22.4	25.9	25.7	24.3	20.5	15.6	11.0		(22)
(23)		5% DB	10.4	11.9	16.0	21.0	25.0	27.9	31.9	33.2	30.2	24.9	19.1	13.4		(23)
(24)		5% MCWB	6.6	8.0	11.4	14.6	17.7	21.5	25.4	25.5	23.7	19.3	15.0	9.4		(24)
(25)		10% DB	9.1	10.4	14.4	19.6	23.8	26.8	30.9	32.3	29.0	23.6	17.7	11.9		(25)
(26)		10% MCWB	5.6	6.7	10.2	13.8	17.2	21.0	25.0	25.2	23.0	18.3	13.8	8.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.4	13.8	15.5	19.0	21.7	25.1	27.3	27.2	26.5	23.8	18.5	14.5		(27)
(28)		0.4% MCDB	12.4	15.4	17.1	21.0	24.0	28.8	32.4	32.7	30.4	26.2	19.9	15.5		(28)
(29)		2% WB	8.7	10.8	13.8	17.5	20.5	24.2	26.6	26.7	25.5	22.0	17.1	11.8		(29)
(30)		2% MCDB	10.3	12.7	16.2	19.8	22.9	27.3	31.4	32.0	29.3	24.9	19.2	13.8		(30)
(31)		5% WB	7.2	8.9	12.3	16.3	19.5	23.5	26.0	26.2	24.9	20.6	15.8	10.2		(31)
(32)		5% MCDB	9.8	11.0	14.8	19.3	22.5	26.2	30.4	31.1	28.4	23.6	18.0	12.4		(32)
(33)		10% WB	5.9	7.2	10.8	15.1	18.6	22.6	25.5	25.8	24.3	19.2	14.5	8.9		(33)
(34)		10% MCDB	8.7	9.7	13.9	18.3	22.0	25.2	29.6	30.4	27.6	22.4	17.0	11.6		(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	8.9	9.7	10.0	9.4	7.5	6.7	7.6	7.9	8.8	9.0	8.6		(35)
(36)		5% DB MCDBR	10.0	10.5	11.3	11.5	11.2	8.9	8.2	8.7	8.8	9.6	9.7	9.9		(36)
(37)		5% WB MCWBR	6.4	7.0	7.2	6.1	5.4	3.9	3.2	3.0	3.7	4.9	5.9	6.4		(37)
(38)		5% WB MCDBR	9.0	8.9	9.4	8.4	8.0	6.4	7.5	7.8	7.1	7.4	8.1	8.7		(38)
(39)	Clear-Sky Solar Irradiance	taub	0.405	0.446	0.509	0.538	0.543	0.565	0.568	0.539	0.490	0.447	0.420	0.399		(40)
(41)		taud	2.127	2.035	1.894	1.877	1.931	1.948	1.997	2.066	2.168	2.200	2.177	2.161		(41)
(42)		Ebn at Noon	767	778	761	763	765	746	741	755	773	772	747	746		(42)
(43)		Edh at Noon	122	149	186	198	190	187	177	162	140	125	115	111		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.20	2.83	3.84	4.78	5.21	4.76	4.97	5.14	3.98	3.24	2.40	1.98		(44)
(45)		RadStd	0.16	0.24	0.30	0.42	0.50	0.42	0.68	0.53	0.37	0.31	0.20	0.14		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page